Aryan Sahni

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Education

Vellore Institute of Technology, Chennai **2023–2027**
B.Tech in Computer Science and Engineering **CGPA: 9.0/10**

Skills

Technical: Python, Java, C/C++, SQL, Git/GitHub, JavaScript, Linux
Frameworks & Libraries: Pandas, NumPy, TensorFlow, LangChain, Scikit-learn, ONNX
AI/ML Expertise: Deep Learning, Computer Vision, NLP, Retrieval-Augmented Generation (RAG), Model Evaluation, Feature Engineering

Summary

Innovative and research-driven Computer Science undergraduate at VIT Chennai with expertise in Artificial Intelligence, Machine Learning, Systems Programming, and Full-Stack Development. Experienced in building AI powered platforms, Graph-RAG systems, and reinforcement learning based Linux process schedulers using TensorFlow, ONNX Runtime, LangChain, and LLM frameworks. Published author at ImageCLEF 2025 in the domain of Medical AI and multimodal image understanding. Skilled in developing scalable, data-driven solutions spanning intelligent automation, kernel level optimization, and retrieval-augmented reasoning systems, with a strong focus on bridging cutting-edge research with real-world applications.

Projects

GMAIL Bait Mail [GitHub]

- Engineered a **Chrome extension** for real-time **phishing and scam detection** leveraging the **Gmail API**, **Flask microservices**, and DOM event tracking via **MutationObserver**.
- Integrated **AWS Bedrock Foundation Models** for automated **LLM-powered threat classification**, contextual **email intent analysis**, and intelligent **response generation**.
- Implemented asynchronous frontend-backend communication pipelines and dynamic content parsing for low-latency inbox monitoring.

AI-Based Process Scheduler [GitHub]

- Developed an **AI-driven Linux process scheduler** using **Reinforcement Learning**, **TensorFlow**, and **ONNX Runtime** for adaptive task prioritization.
- Designed a **dynamic multi-level queue scheduling architecture** optimized for heterogeneous **CPU-bound** and **I/O-bound** workloads.
- Integrated runtime feature extraction pipelines utilizing process metrics such as **utime**, **stime**, memory utilization, and process priority for intelligent scheduling decisions.

Graph-RAG Knowledge Retrieval System [GitHub]

- Architected a **Graph-RAG framework** using **LangChain**, **Ollama**, and **Zephyr LLMs** for scalable **multi-hop semantic retrieval**.
- Integrated **knowledge graph traversal** with **Retrieval-Augmented Generation (RAG)** pipelines to enhance contextual reasoning, entity linking, and response grounding.
- Developed custom evaluation modules for **faithfulness scoring**, **hallucination detection**, **graph coverage analysis**, and **LLM-as-a-judge benchmarking**.
- Optimized containerized deployment workflows and resolved **Docker–Ollama networking** bottlenecks for efficient local inference orchestration.

Publications

Sahni, Aryan, Rachit Gupta, et al. (2025). "Evaluating deep CNNs for multi-label concept detection in ROCOv2 radiology image dataset by team LekshmiscopeVIT". In: *CLEF2025 Working Notes, CEUR Workshop Proceedings, CEUR-WS.org, Madrid, Spain.* URL: https://ceur-ws.org/Vol-4038/paper_203.pdf.

Sahni, Aryan, Krishnesh Mathur, et al. (2026). "An Optimal Evaluation of Entanglement Depth in Variational Quantum Classifiers for Efficient Deployment of Machine Learning Applications". In: *2026 International Conference on Electronic Systems and Intelligent Computing (ICESIC)*, pp. 320–325. DOI: 10.1109/ICESIC67389.2026.11496417.

Achievements

Secured **9th place globally** in the **ImageCLEFmedical 2025** challenge and co-authored a research publication.